

# Fayjullah Haque Emon

Location: Dhaka, Bangladesh

B.Sc. in Computer Science and Engineering

United International University

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GitHub Profile

LinkedIn Profile

## EDUCATION

### •United International University

B.Sc. in Computer Science and Engineering (CGPA: 3.59) | Major: Data Science

Feb. 2022 – Ongoing

Dhaka, Bangladesh

### •Motijheel Boy's Govt. School & College

Higher Secondary Certificate (HSC) – Science (GPA: 5.00)

2018–2020

Dhaka, Bangladesh

### •Sher-E-Bangla School & College

Secondary School Certificate (SSC) – Science (GPA: 5.00)

2016–2018

Dhaka, Bangladesh

## RESEARCH & ACADEMIC WORK

### •Microarchitectural Masking: A JIT-Based Adaptive Dispatcher for Resource-Resilient Serverless Computing[2025]

Academic Research Paper



- Designed and evaluated Microarchitectural Masking, a zero-privilege, user-space JIT dispatcher that successfully neutralizes "Noisy Neighbor" interference in cloud hosts by dynamically shifting to hardware-pressure-optimized algorithmic kernels.
- Ensured virtually no performance loss for compute/IO workloads under contention while recovering ~60% of memory-bound latency.
- Mitigated an estimated 29.6kg of CO<sub>2</sub>e per million invocations seamlessly without requiring underlying infrastructure changes.

### •AI Driven Speech Based Depression Detection System

[2026]

Academic Research Paper



- Implemented machine learning and deep learning architectures including SVM, XGBoost, CNN, RNN, LSTM, and GMM models.
- Extracted robust vocal attributes utilizing openSMILE feature extraction tools and methodologies.
- Evaluated and analyzed system performance using Weighted Average Recall (WAR) and conducted comparative model analysis to optimize predictive reliability.

## PROJECTS

### •Shelf To Tales

[Dates]

Innovative 3D virtual bookshelf for book enthusiasts



- Created an interactive platform featuring a 3D visual bookshelf and personalized reading experiences.
- Integrated backend services to manage book catalogs and user reading history.
- Technologies: Java, Spring Boot, React.js (suggested), HTML/CSS.

### •Smart Cooling Storage (Smart Farm Fresh Storage)

[Dates]

IoT-driven automation system for post-harvest loss reduction



- Built an automated storage system featuring YOLOv8-based vegetable classification and robotic arm sorting.
- Implemented RFID-authenticated dashboards and IoT-driven temperature control via Peltier modules.
- Developed real-time monitoring features for weight sensing and storage conditions.
- Technologies: Python, YOLO, OpenCV, IoT, Arduino/ESP32, RFID.

### •Software Testing Tool

[Dates]

Comprehensive platform for API lifecycle management and test environment automation



- Supports creating multiple environments (Local, Staging) with custom variables utilizing the `{{variable_name}}` syntax framework.
- Automatically extracts and saves JWT authentication tokens to the active environment upon successful system login.
- Enables sending GET, POST, PUT, PATCH, and DELETE requests with custom headers and varied body structures.
- Structures API tests hierarchically into Workspaces, Modules, and timestamped Requests with definitive pass/fail feedback for high traceability.
- Supports exporting and importing the complete test suite as a single unified JSON file to streamline easy team collaboration.

## TECHNICAL SKILLS

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**Languages:** Python, JavaScript, Java, C, HTML, CSS

**Tools:** Figma, Framer Postman, Canva

**Web Technologies:** Spring Boot

**Databases:** MySQL

**Software Quality Assurance:** SQA Fundamentals, API Testing , Junit Testing

**Expertise:** Web Development, AI Research, UI/UX Design

**Soft Skills:** Problem Solving, Team Collaboration, Analytical Thinking, Presentation

## AWARDS & CERTIFICATIONS

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•**5th Runner Up** – UIU CSE Project Show - Microprocessor and Microcontroller Lab

*Summer 2025*

•**Member** – UIU Computer Club (General Member)

*Ongoing*